TED UNIVERSITY

Faculty of Engineering

CMPE 491/SENG491

Senior Project I

LLM Valley

Project Proposal

TEAM 21

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1. Introduction

In this document, we present the proposal for the development of LLM Valley, an AI-driven life simulation demo designed to explore the integration of large language models (LLMs) into interactive NPC systems. The project aims to demonstrate how natural, dynamic conversations can enhance immersion and player engagement in a cozy simulation setting.

The demo will allow players to farm, trade, and build relationships with 4–5 non-playable characters (NPCs) that utilize LLM technology for dynamic dialogue generation. The project's primary objective is to deliver a playable, stable prototype that proves the feasibility of LLM-powered NPCs within a game environment.

The final deliverables will include:

- A working demo for Windows (PC)
- Functional gameplay loop (farm, talk, trade, sleep)
- Integrated LLM-driven dialogue system
- Basic player and NPC stat systems (Love, Trust, Friendship, Mental)
- One complete map with farming and trading features

The development process will progress through three main stages:
requirements development, design and prototyping, and implementation with testing.

2. Project Description

2.1 Overview

LLM Valley is a small scale life simulation demo where the player interacts with a handful of NPCs using natural language input. All NPCs use LLM based sentences to create realistic and varied conversations. Gameplay loop is based on three simple main actions: farming, trading, and building relationships in the village.

Scope of the project includes:

- AI-powered NPC conversation system via Gemini or similar API
- Farming mechanics (planting, watering, harvesting)
- Basic trading system with shopkeeper NPCs
- Simple stat tracking (player and NPC stats)
- Single 2D map with day/night and sleep cycle
- One save slot (save/load system)
- Minimal UI displaying key stats and inventory

Excluded features include multiple areas, animation systems, sound design, or multiplayer — these will be considered for post-demo development.

2.2 Objectives

Objective	Brief Description
Develop a working LLM driven NPC system	Integrate an LLM API to enable dynamic conversations with NPCs.
Build a minimal but complete simulation loop	Implement farming, trading, and relationship mechanics within playable map.
Demonstrate technical feasibility	Showcase that LLM integration can be stable and responsive in a 30 minute gameplay loop.
Achieve smooth and intuitive interaction	Ensure players can communicate with NPCs via keyboard naturally, without tutorials and direct access to API.
Deliver a demo ready prototype	Complete core systems and achieve a stable playthrough

Functional Requirements

- **NPC Dialogue System:** NPCs shall respond dynamically to player input using an integrated LLM API.
- **Stat System:** Player and NPC stats (Love, Trust, Friendship, Mental) shall update according to actions and keywords from player input.
- **Farming System:** The player shall be able to plant, water, and harvest crops across fixed plots.
- **Trading System:** The player shall be able to buy and sell goods with shopkeeper NPCs.

- **Save/Load System:** The game shall store player progress in one save slot.
- **Day/Night System:** Sleeping shall advance the in-game day and restore the player's Mental stat.

Non-Functional Requirements

- **Performance:** LLM responses shall appear within 3 seconds for 90% of inputs.
- **Usability:** The interface shall allow intuitive interaction without tutorial requirements and direct intervention.
- **Reliability:** The demo shall run without crashes for a full in-game cycle.
- **Maintainability:** The LLM system shall use modular prompts to allow easy updates.
- **Cost Efficiency:** Monthly LLM usage shall stay within a predetermined budget cap from API.

3. Target Customers

1. Narrative Gamers
Players who enjoy good quality narrative games and seek for character interaction that leads to roleplay feeling

2. AI Researcher Game Developers:
LLM implementation in object oriented game programming is a very niche topic that will hopefully target a community of cutomers which are game developers that are also interested in AI research and implementation.

3. Investors:
Stakeholders that will be interested in investing such projects. We hope these customers to generate a flow in the industry which will then lead to more projects with actual llm implementation.

4. Value Proposition

1. **AI Driven Story Telling:**
   LLM valley shows a new innovative way of implementing advanced ai systems and llms being implemented in object orianted software and games.
2. **Proof of Concept for LLM Integration:**
   The demo provides a technically verified framework for integrating large language models into small-scale interactive simulations.
3. **Accessible and Scalable Design:**
   With its modular, minimal systems, the game can be easily expanded with more NPCs, areas, or mechanics if the demo succeeds.
4. **Educational and Research Value:**
   The project serves as a practical example for game developers and academic studies focusing on human-AI interaction.

5. Preliminary Plan

5.1 Planned Deliverables

Phase 1: Requirements Development and Analysis

- Outcome: Detailed system breakdown and simplified GDD (already completed).

Phase 2: Design and Prototyping

- Outcome:
  UI mockups and game flow diagrams
  Prototype map with placeholder assets
  Initial LLM API connection test

Phase 3: Implementation and Testing

- Outcome:
  Fully functional demo with farming, trading, and NPC dialogue
  Keyword-based stat tracking system
  Testing reports ensuring system stability and stat balance

5.2 Working Plan

Development Methodology:
We will work with a plan-driven, agile methodology.

- The plan-driven approach ensures all base systems are implemented in early stages of the project.
- The agile elements (weekly iterations and feedback testing) allow flexibility for tuning LLM behavior and player feedback.

Phases:

1. Requirements Development and Analysis (Week 1–2)
   a. Finalize technical needs for LLM integration
   b. Define internal data structures and gameplay flow
2. Design and Prototyping (Week 3–4)
   a. Implement player movement, farming UI, and basic NPC mockup
   b. Integrate first working LLM dialogue
3. Development and Testing (Week 5–12)
   a. Complete systems for stats, trading, and progression
   b. Conduct internal playtests and adjust balance
   c. Perform bug fixing, polish UI, and prepare demo build

5.3 Development Team

- Project Manager: Oversees scope, timeline, and deliverables
- Programmer / LLM Engineer: Implements core logic and API integration
- UI/UX Designer: Creates interface mockups and visual flow
- 2D Artist / Technical Designer: Manages map layout and assets integration

5.4 Training and Collaboration

- Training: Team members will learn to optimize prompt engineering for stable LLM responses.
- Collaboration Platform: Development coordination will mainly be handled through weekly face-to-face scrum meetings. MS Teams and Github will be used for online collaboration, sprint reviews, and file sharing.

Meetings:

- Daily stand-ups (progress & blockers)
- Weekly planning and task review meetings
- Bi-weekly playtest sessions and demo feedback reviews

5.5 Milestones

1. Completion of core movement, farming, and UI systems (End of Month 1)
2. Completion of 4 NPCs and working stat system (Mid Month 2)
3. Final integration and stable demo build (End of Month 3)

6. Conclusion

This proposal outlines the structured plan to develop *LLM Valley*, a compact but innovative life simulation demo demonstrating LLM-driven NPCs in action. The project's success will validate the potential of integrating generative AI into interactive gameplay and narrative systems, setting the foundation for future, more expansive projects.

7. References

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